

3 Relativistic Quantum Mechanics.

3.1. Relativistic wave function for Spin-0 particles.

The Klein-Gordon Equation.

The description of phenomena at high energies requires the ~~inner~~ investigation of relativistic wave equation, which means equations which are invariant under Lorentz transformations.

There are several concepts

1). spatial and temporal coordinates have to be treated equally within the theory.

2). $\Delta x \sim \frac{\hbar}{\Delta p} \sim \frac{\hbar}{m_0 c}$, so a relativistic particle ~~ca~~ cannot be localized more accurately than $\sim \frac{\hbar}{m_0 c}$. If you try to squeeze a single particle into a space smaller than a certain limit, the uncertainty principle cause its energy to fluctuate so widely that the extra energy to literally create new particle-antiparticle pairs out of the vacuum. Therefore, you can only think of a particle as a single, isolated "free particle" if you don't try to measure its location more precisely than its Compton wavelength $\lambda_c = \frac{\hbar}{m_0 c}$.

3). The time is also uncertain $\Delta t \sim \frac{\Delta x}{c} > \frac{\hbar}{m_0 c^2}$. Because of the finite speed of light, a spatial smear guarantees a temporal smear. You cannot define a perfectly "fixed time t " across the entire spread of the particle.

4.) At relativistic energies particle conservation is no longer a valid assumption. A relativistic theory must be able to describe pair creation, vacuum polarization, particle conversion, et.

First we shall remark on the notation used metric tensor:

$$g_{\mu\nu} = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix} \text{ and } g^{\mu\nu} = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}$$

contravariant four-vector:

$$\begin{cases} x^\mu = \{ct, x, y, z\} \\ x_\mu = \{ct, -x, -y, -z\} \end{cases} \quad /$$

the indices can be "raised" to give:

$$x^\mu = g^{\mu\nu} x_\nu$$

four-momentum vector:

$$p^\mu = \left\{ \frac{E}{c}, p_x, p_y, p_z \right\}$$

four-momentum operator:

$$\hat{p}^\mu = i\hbar \frac{\partial}{\partial x_\mu} \equiv i\hbar \nabla^\mu = i\hbar \left\{ \frac{\partial}{\partial ct}, -\frac{\partial}{\partial x}, -\frac{\partial}{\partial y}, -\frac{\partial}{\partial z} \right\}$$

the commutation relations of momentum and position:

$$\begin{aligned} [\hat{p}^\mu, x^\nu] \psi &= i\hbar \left[\frac{\partial}{\partial x_\mu}, g^{\nu\sigma} x_\sigma \right] \psi \\ &= i\hbar \left[\frac{\partial}{\partial x_\mu} g^{\nu\sigma} x_\sigma - g^{\nu\sigma} x_\sigma \frac{\partial}{\partial x_\mu} \right] \psi \end{aligned}$$

$$= i\hbar \left(g^{\nu\sigma} x_\sigma \frac{\partial \psi}{\partial x_\mu} + g^{\nu\sigma} \psi \cdot \frac{\partial x_\sigma}{\partial x_\mu} - g^{\nu\sigma} x_\sigma \frac{\partial \psi}{\partial x_\mu} \right)$$

$$= i\hbar g^{\nu\sigma} \frac{\partial x_\sigma}{\partial x_\mu} \cdot \psi$$

$$= i\hbar g^{\nu\sigma} \delta_\sigma^\mu \cdot \psi$$

$$= i\hbar g^{\nu\mu} \psi$$

so we can get:

$$[\hat{p}^\mu, x^\nu] = i\hbar g^{\nu\mu} = i\hbar g^{\mu\nu}$$

the four-potential of electromagnetic field is given by

$$A^\mu = \{A_0, A_x, A_y, A_z\}$$

the electromagnetic field tensor follows in the following way:

$$F^{\mu\nu} = \partial^\nu A^\mu - \partial^\mu A^\nu = \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{pmatrix} \quad (c=1)$$

The Klein-Gordon Equation

The Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} \psi = \left[-\frac{\hbar^2}{2m_0} \nabla^2 + V(x) \right] \psi$$

corresponds to the nonrelativistic energy relation in operator form:

$$\hat{E} = \frac{\hat{p}^2}{2m_0} + V(x) \quad \text{and} \quad \hat{E} = i\hbar \frac{\partial}{\partial t}$$

In order to obtain a relativistic wave function we start by considering free particles with the relativistic relation:

$$E^2 = (cp)^2 + (m_0c)^2.$$

$$\Rightarrow p^\mu p_\mu = (m_0c)^2$$

~~Thus~~ We replace the four-momentum p^μ by the four-momentum operator and get the Klein-Gordon equation for free particles:

$$\hat{p}^\mu \hat{p}_\mu \psi = m_0^2 c^2 \psi$$

Here m_0 is the rest mass of the particle. We can rewrite it as:

$$\left(\frac{\partial^2}{c^2 \partial t^2} - \frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2} \right) \psi = \frac{m_0^2 c^2}{\hbar^2} \psi.$$

This is similar to classical wave equation, so the free solutions are of the form:

~~$$\psi = \exp\left(\frac{i(m_0c)}{\hbar}(ct)\right)$$~~

$$\psi = \exp\left(\frac{i}{\hbar} p_\mu x^\mu\right) = \exp\left[-\frac{i}{\hbar}(Et - \vec{p} \cdot \vec{x})\right]$$

So we have:

$$\hat{p}_\mu \hat{p}^\mu \exp\left(\frac{i}{\hbar} p_\mu x^\mu\right) = m_0^2 c^2 \psi$$

$$\Rightarrow p^\mu p_\mu \psi = m_0^2 c^2 \psi$$

$$\Rightarrow E^2 = m_0^2 c^2 + (\cancel{cp})^2 c^2 \vec{p}^2.$$

which results in:

$$E = \pm (m_0^2 c^2 + c \vec{p}^2)^{\frac{1}{2}}.$$

The negative energy are physically connected with antiparticles

Next we construct the four-current j_μ .

$$\begin{cases} (\vec{p}_\mu \vec{p}^\mu - m_0^2 c^2) \psi = 0 & (1) \\ (\vec{p}_\mu \vec{p}^\mu - m_0^2 c^2) \psi^* = 0 & (2) \end{cases}$$

We get $(\psi^* (1) - \psi (2))$:

$$\begin{aligned} & \psi^* (\vec{p}_\mu \vec{p}^\mu - m_0^2 c^2) \psi - \psi (\vec{p}_\mu \vec{p}^\mu - m_0^2 c^2) \psi^* \\ &= \psi^* \vec{p}_\mu \vec{p}^\mu \psi - \psi \vec{p}_\mu \vec{p}^\mu \psi^* \\ &= -\hbar^2 \psi^* \nabla_\mu \nabla^\mu \psi + \hbar^2 \psi \nabla_\mu \nabla^\mu \psi^* \\ &= \hbar^2 (\psi \nabla_\mu \nabla^\mu \psi^* - \psi^* \nabla_\mu \nabla^\mu \psi) \\ &= 0 \end{aligned}$$

we also have :

$$\begin{cases} \nabla_\mu (\psi \nabla^\mu \psi^*) = (\nabla_\mu \psi) (\nabla^\mu \psi^*) + \psi (\nabla_\mu \nabla^\mu \psi^*) \\ \nabla_\mu (\psi^* \nabla^\mu \psi) = (\nabla_\mu \psi^*) (\nabla^\mu \psi) + \psi^* (\nabla_\mu \nabla^\mu \psi) \\ (\nabla_\mu \psi) (\nabla^\mu \psi^*) = (\nabla_\mu \psi^*) (\nabla^\mu \psi) \end{cases}$$

So we obtain :

$$\begin{aligned}
& \psi \partial_\mu \partial^\mu \psi^* - \psi^* \partial_\mu \partial^\mu \psi \\
&= \partial_\mu (\psi \partial^\mu \psi^* - \psi^* \partial^\mu \psi) \\
&= \frac{2m_0}{i\hbar} \partial_\mu j^\mu
\end{aligned}$$

$$= 0$$

where four-current density is

$$\begin{aligned}
j^\mu &= \frac{i\hbar}{2m_0} (\psi \partial^\mu \psi^* - \psi^* \partial^\mu \psi) \\
&= \frac{i\hbar}{2m_0} \left\{ \frac{1}{c} (\psi \partial_t \psi^* - \psi^* \partial_t \psi), -(\psi \nabla \psi^* - \psi^* \nabla \psi) \right\}
\end{aligned}$$

So we have:

$$\begin{aligned}
\partial_\mu j^\mu &= \frac{\partial}{\partial t} \left[\frac{i\hbar}{2m_0 c} (\psi \partial_t \psi^* - \psi^* \partial_t \psi) \right] + \nabla \cdot \left[\frac{i\hbar}{2m_0} (\psi \nabla \psi^* - \psi^* \nabla \psi) \right] \\
&= \frac{\partial}{\partial t} \rho + \nabla \cdot \vec{j} \\
&= 0
\end{aligned}$$

This is the form of a continuity equation. As usual, integration over the entire configuration space yields:

$$\int_V \frac{\partial \rho}{\partial t} d^3x = \frac{\partial}{\partial t} \int_V \rho d^3x = - \int_V \nabla \cdot \vec{j} d^3x = - \int_{\partial V} \vec{j} \cdot d\vec{S} = 0$$

Hence,

$$\int_V \rho d^3x = \text{const.}$$

So it would be a natural guess to interpret ρ as a probability density.

$$\rho = \frac{i\hbar}{2m_0 c} (\psi \partial_t \psi^* - \psi^* \partial_t \psi)$$

3.2 A Wave Equation for spin- $\frac{1}{2}$ particles - The Dirac Equation

The Schrödinger equation is linear in the time derivative, it is natural to try to construct a Hamiltonian that is also linear in the spation derivatives. (equality of spatial and temporal coordinates). So the desired equation has to be of the form:

$$i\hbar \frac{\partial}{\partial t} \psi = \left[-i\hbar c \cdot (\vec{\alpha}_1 \cdot \frac{\partial}{\partial x_1} + \vec{\alpha}_2 \frac{\partial}{\partial x^2} + \vec{\alpha}_3 \frac{\partial}{\partial x^3}) + \beta m_0 c^2 \right] \psi \quad (1)$$

We suspect that the $\vec{\alpha}_i$ are matrices and ψ has to be a column vector.

$$\psi = \begin{pmatrix} \psi_1(x,t) \\ \vdots \\ \psi_N(x,t) \end{pmatrix}$$

We shall call them spinor and assume their dimension is N . So we have:

$$i\hbar \frac{\partial}{\partial t} \psi_0 = -i\hbar c \cdot \sum_{i=1}^N \left[-i\hbar c (\vec{\alpha}_i \cdot \vec{p}_i \right.$$

$$\left. + m_0 c^2 \beta_{0i} \right] \psi_i$$

To fulfill the correct energy-momentum relation for relativistic free particle:

$$E^2 = (\vec{p}c)^2 + (m_0 c^2)^2$$

every single component ψ_0 of the spinor ψ has to satisfy the Klein-Gordon equation:

$$-\hbar^2 \frac{\partial^2}{\partial t^2} \psi_0 = (-\hbar^2 c^2 \nabla^2 + m_0^2 c^4) \psi_0$$

From equation (1), we have:

$$\begin{aligned}
 -\hbar^2 \frac{\partial^2}{\partial t^2} \psi &= (c \cdot \sum_i \vec{\alpha}_i \vec{p}_i + m_0 c^2 \beta) \cdot \psi (c \sum_j \vec{\alpha}_j \vec{p}_j + m_0 c^2 \beta) \psi \\
 &= [c^2 \sum_i \vec{\alpha}_i^2 \vec{p}_i^2 + c^2 \sum_{i \neq j} (\vec{\alpha}_i \vec{\alpha}_j \vec{p}_i \vec{p}_j + \vec{\alpha}_j \vec{\alpha}_i \vec{p}_j \vec{p}_i) + m_0 c^2 \sum_i (\vec{\alpha}_i \beta + \beta \vec{\alpha}_i) \vec{p}_i + \beta^2 (m_0 c^2)^2 \vec{p}^2] \psi \\
 &\quad = (\sum_i \vec{\alpha}_i^2 \vec{p}_i^2 + c^2 \sum_{i \neq j} (\vec{\alpha}_i \vec{\alpha}_j + \vec{\alpha}_j \vec{\alpha}_i) \vec{p}_i \vec{p}_j + m_0 c^2 \sum_i (\vec{\alpha}_i \beta + \beta \vec{\alpha}_i) \vec{p}_i + \beta^2 (m_0 c^2)^2 \vec{p}^2) \psi
 \end{aligned}$$

Comparison with Klein-Gordon equation shows the following requirements for the matrices $\vec{\alpha}_i, \beta$:

$$\begin{cases}
 \vec{\alpha}_i^2 = \mathbb{I}, \quad \vec{\alpha}_i = \vec{\alpha}_i^{-1} \\
 \vec{\alpha}_i \vec{\alpha}_j + \vec{\alpha}_j \vec{\alpha}_i = 0, \quad (i \neq j) \\
 \vec{\alpha}_i \beta + \beta \vec{\alpha}_i = 0 \\
 \beta^2 = \mathbb{I}, \quad \beta = \beta^{-1}
 \end{cases}$$

In order to establish hermiticity of the Hamiltonian $\hat{H}$, the matrices $\vec{\alpha}_i, \beta$ also have to be Hermitian:

$$\vec{\alpha}_i = \vec{\alpha}_i^\dagger \quad \text{and} \quad \beta = \beta^\dagger$$

Therefore, the eigenvalues of the matrices are real and can only have the values ± 1 . We assume $\vec{\alpha}_i$ in its $\hat{A}_i$ which is the ~~eigenre-~~ eigenrepresentation of $\hat{A}_i$ has the form:

$$\hat{U}^\dagger \hat{A}_i \hat{U} = \hat{A}_i = \begin{pmatrix} A_1 & 0 & 0 & \dots & 0 \\ 0 & A_2 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & A_N \end{pmatrix}$$

with the eigenvalues $A_1, \dots, A_N$ and $A_j^2 = 1, (j=1 \dots N)$. So $A_j = \pm 1$

Furthermore, from the anticommutation relation ($\hat{\alpha}_i \hat{\beta} + \hat{\beta} \hat{\alpha}_i = 0$) it follows that the trace of ~~the~~ each $\hat{\alpha}_i$ and $\hat{\beta}$ has to be zero,

$$\begin{aligned} \text{tr}(\hat{\alpha}_i) &= \text{tr}(\hat{1} \hat{\alpha}_i) \\ &= \text{tr}(\hat{\beta}^2 \hat{\alpha}_i) \quad (\hat{\beta}^2 \hat{\alpha}_i = \hat{\beta} \cdot \hat{\beta} \hat{\alpha}_i, \text{tr} AB = \text{tr} BA) \\ &= \text{tr}(\hat{\beta} \hat{\alpha}_i \hat{\beta}) \quad (\hat{\alpha}_i \hat{\beta} + \hat{\beta} \hat{\alpha}_i = 0 \Rightarrow \hat{\alpha}_i = -\hat{\beta} \hat{\alpha}_i \hat{\beta} = -\hat{\beta} \hat{\alpha}_i \hat{\beta}) \\ &= -\text{tr} \hat{\alpha}_i \end{aligned}$$

$$\Rightarrow \text{tr}(\hat{\alpha}_i) = 0.$$

Similarly, $\text{tr}(\hat{\beta}) = 0$. So we can get:

$$\text{tr}(\hat{A}_i) = \text{tr}(U^\dagger \hat{\alpha}_i U) = \text{tr}(\hat{\alpha}_i) = 0$$

$$\Rightarrow \sum_k A_k = 0.$$

Because the eigenvalues of $\hat{\alpha}_i$ and $\hat{\beta}$ are equal to ± 1 , each matrix $\hat{\alpha}_i$ and $\hat{\beta}$ has to possess as many positive as negative eigenvalues. and therefore has to be of even dimension. The smallest even dimension ($N=2$) can not be right because only three anticommuting matrices exist (Pauli matrices $\hat{\sigma}_i$). Therefore the smallest dimension ~~for~~ is four. One possible explicit representation of the Dirac matrices are:

$$\hat{\alpha}_i = \begin{pmatrix} 0 & \hat{\sigma}_i \\ \hat{\sigma}_i & 0 \end{pmatrix} \quad \text{and} \quad \hat{\beta} = \begin{pmatrix} \hat{1} & 0 \\ 0 & -\hat{1} \end{pmatrix}$$

where $\hat{\sigma}_i$ are Pauli's 2×2 matrices, and $\hat{1}$ is the unit matrix

Next we construct the four-current density and the equation of continuity. From Dirac ~~equation~~ equation and $\psi^\dagger$ we obtain:

$$i\hbar \psi^\dagger \frac{\partial}{\partial t} \psi = \psi^\dagger \cancel{c} (\vec{\alpha} \cdot c \vec{p}) \psi + mc^2 \psi^\dagger \beta \psi \quad (1)$$

Furthermore, we form the Hermitian conjugate of Dirac equation:

$$-i\hbar \frac{\partial}{\partial t} \psi^\dagger = \psi^\dagger (\vec{\alpha} \cdot c \vec{p})^\dagger + mc^2 \psi^\dagger \beta^\dagger$$

and multiply this equation from the right by ψ , to give:

$$-i\hbar \left(\frac{\partial}{\partial t} \psi^\dagger \right) \psi = \psi^\dagger (\vec{\alpha} \cdot c \vec{p})^\dagger \psi + mc^2 \psi^\dagger \beta^\dagger \psi \quad (2)$$

Taking into consideration the hermiticity of the Dirac matrices ($\alpha_i^\dagger = \alpha_i$, $\beta^\dagger = \beta$), subtraction of (2) from (1) yields:

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} (\psi^\dagger \psi) &= \psi^\dagger (\vec{\alpha} \cdot c \vec{p}) \psi - \psi^\dagger (\vec{\alpha} \cdot c \vec{p})^\dagger \psi \\ &= -i\hbar (\psi^\dagger (\vec{\alpha} \cdot c \nabla) \psi) - i\hbar \psi^\dagger (\vec{\alpha}^\dagger \cdot c \nabla) \psi \\ &= -i\hbar c [\psi^\dagger (\vec{\alpha} \cdot \vec{\nabla}) \psi + \psi^\dagger (\vec{\alpha} \cdot \vec{\nabla}) \psi] \\ &= -i\hbar c \cdot \nabla \cdot \psi^\dagger \vec{\alpha} \psi \end{aligned}$$

So we can get:

$$\frac{\partial}{\partial t} \rho + \nabla \cdot \vec{j} = 0$$

where $\rho = \psi^\dagger \psi$ and $\vec{j} = c \psi^\dagger \vec{\alpha} \psi$

ρ is positive definite and because of the conservation law

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{j} = 0$$

we can accept the ~~interpret~~ interpretation of ρ as a probability density. Accordingly, we call $\vec{j}$ the probability current density.

So $\{\rho, \vec{j}\}$ should form a four-vector:

$$j^\mu = \{\rho, \vec{j}\}$$

3.2. | Free Motion of a Dirac Particle.

The free Dirac equation:

$$i\hbar \frac{\partial \psi}{\partial t} = (c \vec{\alpha} \cdot \vec{p} + mc^2 \beta) \psi$$

Its stationary states are found with ansatz:

$$\psi(\vec{x}, t) = \psi(\vec{x}) \exp\left[-\frac{iEt}{\hbar}\right],$$

~~wh~~ with which transforms free Dirac equation into:

$$E \psi(\vec{x}) = (c \vec{\alpha} \cdot \vec{p} + mc^2 \beta) \psi(\vec{x})$$

It is useful to split up the four-component spinor into ~~two~~ two component spinors ϕ and χ :

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix} = \begin{pmatrix} \phi \\ \chi \end{pmatrix} \quad \text{with} \quad \begin{cases} \phi = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} \\ \chi = \begin{pmatrix} \psi_3 \\ \psi_4 \end{pmatrix} \end{cases}$$

we can obtain:

$$\mathcal{E} \begin{pmatrix} \psi \\ \chi \end{pmatrix} = c \begin{pmatrix} 0 & \vec{\sigma} \\ \vec{\sigma} & 0 \end{pmatrix} \cdot \vec{p} \begin{pmatrix} \psi \\ \chi \end{pmatrix} + mc^2 \begin{pmatrix} \mathbb{I} & 0 \\ 0 & -\mathbb{I} \end{pmatrix} \begin{pmatrix} \psi \\ \chi \end{pmatrix}$$

$$\Rightarrow \begin{cases} \mathcal{E}\psi = c \vec{\sigma} \cdot \vec{p} \chi + mc^2 \psi \\ \mathcal{E}\chi = c \vec{\sigma} \cdot \vec{p} \psi - mc^2 \chi \end{cases}$$

States with definite momentum $\vec{p}$ are.

$$\begin{pmatrix} \psi \\ \chi \end{pmatrix} = \begin{pmatrix} \psi_0 \\ \chi_0 \end{pmatrix} \exp\left[\frac{i \vec{p} \cdot \vec{x}}{\hbar}\right]$$

So we can get: ($\vec{p}$ is the eigenvalues of $\vec{p}$)

$$\begin{cases} (\mathcal{E} - mc^2) \psi_0 - c \vec{\sigma} \cdot \vec{p} \chi_0 = 0 \\ -c \vec{\sigma} \cdot \vec{p} \psi_0 + (\mathcal{E} + mc^2) \chi_0 = 0 \end{cases}$$

This equation has nontrivial solutions only in the case of a vanishing determinant of the coefficients:

$$\begin{vmatrix} (\mathcal{E} - mc^2) \mathbb{I} & -c \vec{\sigma} \cdot \vec{p} \\ -c \vec{\sigma} \cdot \vec{p} & (\mathcal{E} + mc^2) \mathbb{I} \end{vmatrix} = 0$$

using the relation:

$$(i \vec{\sigma} \cdot \vec{A})(i \vec{\sigma} \cdot \vec{B}) = \vec{A} \cdot \vec{B} \mathbb{I} + i \vec{\sigma} \cdot (\vec{A} \times \vec{B})$$

we have:

$$\begin{aligned} [\mathcal{E}^2 - (mc^2)^2] \mathbb{I} &= c^2 (\vec{\sigma} \cdot \vec{p})^2 \\ &= c^2 \vec{p}^2 \end{aligned}$$

Finally, we can get:

$$\mathcal{E} = \pm E_p \quad \text{with} \quad E_p = c \sqrt{p^2 + m_0^2 c^2}$$

From:

$$-c \vec{\sigma} \cdot \vec{p} \psi_0 + (\mathcal{E} + m_0 c^2) \chi_0 = 0$$

for fixed $\mathcal{E}$, we can get:

$$\chi_0 = \frac{c (\vec{\sigma} \cdot \vec{p})}{m_0 c^2 + \mathcal{E}} \psi_0$$

let's us denote the two spinor ψ_0 in the form.

$$\psi_0 = U = \begin{pmatrix} U_1 \\ U_2 \end{pmatrix} \quad \text{with} \quad U^\dagger U = \mathbb{I}.$$

We obtain the complete set of positive and negative free solutions of the Dirac equation as:

$$\bar{\Psi}_{\vec{p}, \lambda}(\vec{x}, t) = N \cdot \left(\frac{c (\vec{\sigma} \cdot \vec{p})}{m_0 c^2 + \lambda E_p} U \right) \frac{\exp\left[\frac{i}{\hbar} (\vec{p} \cdot \vec{x} - \lambda E_p t)\right]}{(2\pi \hbar)^{\frac{3}{2}}}.$$

Here $\lambda = \pm 1$ characterizes the positive and negative solutions with time evolution factor $\mathcal{E} = \lambda E_p$, and N is the normalization factor. So we have:

$$\int \bar{\Psi}_{\vec{p}, \lambda}^\dagger \bar{\Psi}_{\vec{p}', \lambda'} d^3 x = \delta_{\lambda \lambda'} \delta(\vec{p} - \vec{p}')$$

Hence,

$$N^2 \left[U^\dagger U + U^\dagger \frac{c^2 (\vec{\sigma} \cdot \vec{p})^2}{(m_0 c^2 + \lambda E_p)^2} U \right] = 1$$

using the relation:

$$(\vec{\sigma} \cdot \vec{p})^2 = (\vec{p})^2 \cdot \mathbb{I},$$

we have

$$N^2 \left[1 + \frac{c^2 p^2}{(m_0 c^2 + \lambda E_p)^2} \right] = 1$$

$$\Rightarrow N = \sqrt{\frac{(m_0 c^2 + \lambda E_p)^2}{c^2 p^2 + (m_0 c^2 + \lambda E_p)^2}}$$

$$= \sqrt{\frac{(m_0 c^2 + \lambda E_p)^2}{c^2 p^2 + (m_0 c^2)^2 + 2\lambda E_p (m_0 c^2) + (\lambda E_p)^2}} \quad [c^2 p^2 + (m_0 c^2)^2 = (E_p)^2]$$

$$= \sqrt{\frac{(m_0 c^2 + \lambda E_p)^2}{2\lambda E_p (m_0 c^2 + \lambda E_p)}}$$

$$= \sqrt{\frac{(m_0 c^2 + \lambda E_p)}{2\lambda E_p}}$$

For every momentum $\vec{p}$, there are two different kinds of solutions,

We will now need another quantum number because the energy eigenvalues are two-fold ~~degeneracy~~ degenerate. It is

helicity. First we note that the operator:

$$\vec{\Sigma} \cdot \vec{p} = \begin{pmatrix} \vec{\sigma} & 0 \\ 0 & \vec{\sigma} \end{pmatrix} \cdot \vec{p}$$

and the four-dimensional generalization of ~~of~~ the spin vector operator:

$$\vec{S} = \frac{\hbar}{2} \cdot \vec{\Sigma}$$

We calculate

$$[\hat{H}_f, \vec{\Sigma} \cdot \vec{p}] = [c \vec{\alpha} \cdot \vec{p} + mc^2 \beta, \vec{\Sigma} \cdot \vec{p}]$$

$$= [c \vec{\alpha} \cdot \vec{p}, \vec{\Sigma} \cdot \vec{p}] + mc^2 [\beta, \vec{\Sigma} \cdot \vec{p}]$$

Furthermore, we obtain: ~~$(\vec{\alpha} \cdot \vec{p}, \vec{\Sigma} \cdot \vec{p}) = 0$~~

$$[\vec{\alpha} \cdot \vec{p}, \vec{\Sigma} \cdot \vec{p}] \stackrel{\text{notation}}{=} \alpha \cdot p \cdot \Sigma \cdot p - \Sigma \cdot p \cdot \alpha \cdot p \quad (\text{simplified})$$

$$= \begin{pmatrix} 0 & \sigma \cdot p \\ \sigma \cdot p & 0 \end{pmatrix} \begin{pmatrix} \sigma \cdot p & 0 \\ 0 & \sigma \cdot p \end{pmatrix} - \begin{pmatrix} \sigma \cdot p & 0 \\ 0 & \sigma \cdot p \end{pmatrix} \begin{pmatrix} 0 & \sigma \cdot p \\ \sigma \cdot p & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & (\sigma \cdot p)^2 \\ (\sigma \cdot p)^2 & 0 \end{pmatrix} - \begin{pmatrix} 0 & (\sigma \cdot p)^2 \\ (\sigma \cdot p)^2 & 0 \end{pmatrix}$$

$$= 0$$

Hence:

$$[\hat{H}_f, \vec{\Sigma} \cdot \vec{p}] = 0$$

and naturally

$$[\vec{p}, \vec{\Sigma} \cdot \vec{p}] = p \begin{pmatrix} \sigma \cdot p & 0 \\ 0 & \sigma \cdot p \end{pmatrix} - \begin{pmatrix} \sigma \cdot p & 0 \\ 0 & \sigma \cdot p \end{pmatrix} p$$

$$= 0$$

This means that $\vec{\Sigma} \cdot \vec{p}$, $\hat{H}_f$ and $\vec{p}$ can be diagonalized together

The same holds for the helicity operator

$$\hat{\Lambda}_S = \frac{\hbar}{2} \cdot \vec{\Sigma} \cdot \frac{\vec{p}}{|\vec{p}|} = \vec{S} \cdot \frac{\vec{p}}{|\vec{p}|}$$

If the electron wave propagates into the direction of z axis, we have:

$$\vec{p} = \{0, 0, p\}$$

so $\hat{\Lambda}_S$ becomes:

$$\hat{\Lambda}_S = \hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} \hat{\sigma}_z & 0 \\ 0 & \hat{\sigma}_z \end{pmatrix}$$

the eigenvectors of $\hat{\Lambda}_z$ are:

$$\lambda = \frac{\hbar}{2} \begin{cases} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{cases}, \quad \lambda = -\frac{\hbar}{2} \begin{cases} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{cases}$$

we denote Dirac wave by

$$\begin{cases} \bar{\Psi}_{p, \lambda, +\frac{1}{2}} = N \begin{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \frac{c\hat{\sigma}_z p}{mc^2 + \lambda E_p} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} \exp \left[\frac{i}{\hbar} (p z - \lambda E_p t) \right] \\ \bar{\Psi}_{p, \lambda, -\frac{1}{2}} = N \begin{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ \frac{c\hat{\sigma}_z p}{mc^2 + \lambda E_p} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} \exp \left[\frac{i}{\hbar} (p z - \lambda E_p t) \right] \end{cases}$$